# 🚀 MERN Portfolio Builder

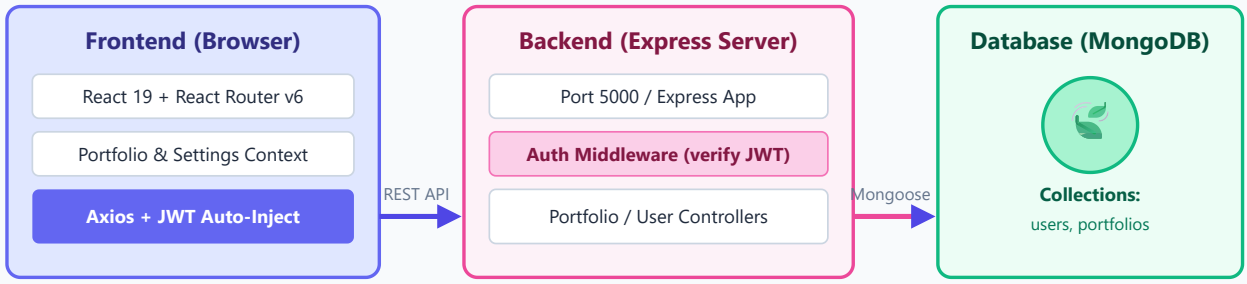
Complete Architecture, Flowcharts, Lifecycle & Step-by-Step Technical Documentation

React 19 & Vite | Node.js & Express | MongoDB | Tailwind v4 & DaisyUI v5 | JWT Auth

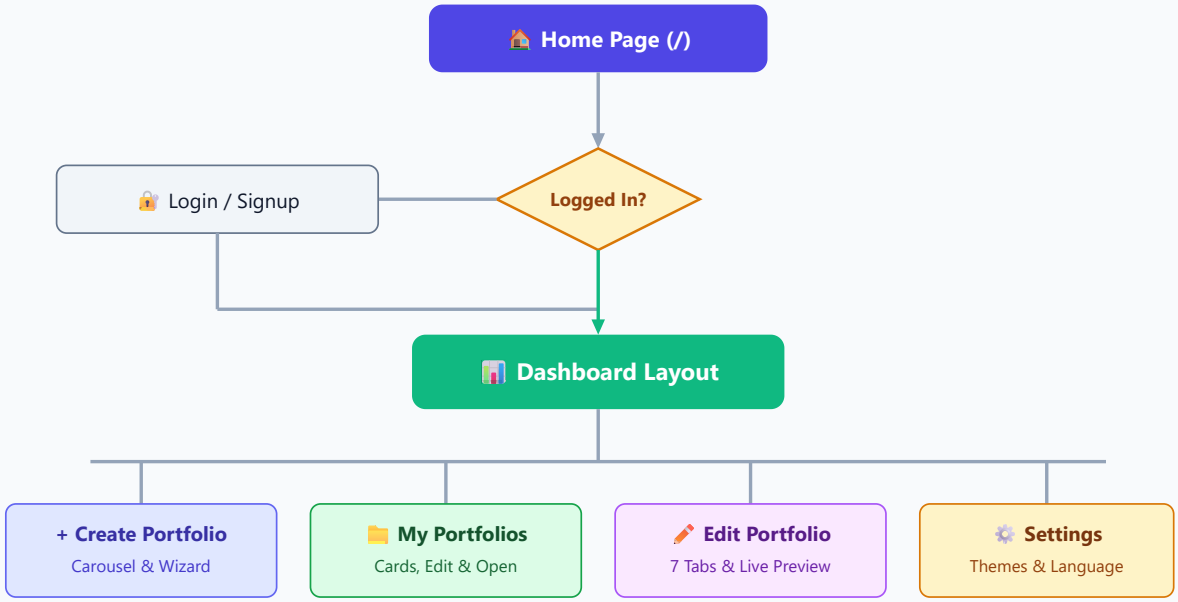
## 📌 1. Project Overview (Poora Project Kya Karta Hai)

Yeh application ek **Full-Stack MERN Portfolio Management System** hai jahan software developers, designers aur students aasani se bina coding kiye apna clean, customized aur responsive portfolio website bana kar publish kar sakte hain. Isme auto-saved section forms, canvas-based image resizing, DaisyUI themes switcher, live interactive preview, aur **Public vs Private security control** diya gaya hai.

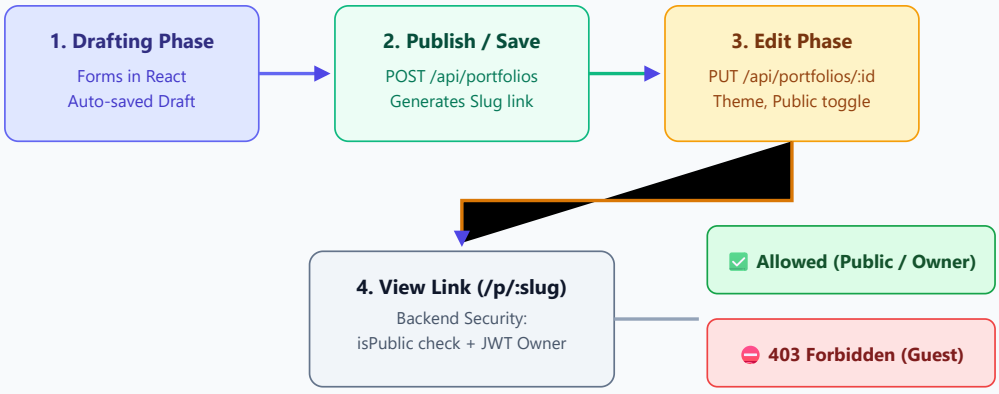
## 📊 2. High-Level System Architecture Flowchart



## 🗺️ 3. User Journey & Navigation Flowchart




## 🔄 4. Portfolio Lifecycle & Privacy Verification Flowchart



## 📖 5. Step-by-Step Detailed Written Flow

**Step 1: User Signup, Login & JWT Handling**

- User jab `/signup` par jata hai, toh apna naam, email aur password fill karta hai.
- Backend password ko `bcrypt.hash()` karke MongoDB mein save karta hai aur JWT token issue karta hai.
- User login hone ke baad token `localStorage.setItem('token', token)` mein store hota hai.
- Axios Interceptor har request ke saath automatically `Authorization: Bearer <token>` attach karta hai.

**Step 2: Dashboard Overview (/dashboard)**

- Logged-in user ke sabhi portfolios database se fetch hote hain ( `GET /api/portfolios` ).
- Stat cards mein: Total Portfolios, Public vs Private count, Total Views aur Average Profile Completeness dikhti hai.
- Quick actions ke zariye user turant continue editing kar sakta hai ya new portfolio bana sakta hai.

**Step 3: Portfolio Creation (/create-portfolio & /create-portfolio/:section)**

- 6 sections ka visual carousel milta hai: Basic Info, Contact, Skills, Projects, Experience, aur Education.
- Profile image upload karne par HTML5 Canvas ke zariye image 256x256 square crop ho kar base64 mein save hoti hai.
- Poora form `PortfolioContext` ke through local browser storage mein safe rehta hai.
- Publish button dabane par `POST /api/portfolios` request call hoti hai, jo user ke naam se readable slug link create karke database mein save kar deti hai.

**Step 4: Portfolio Edit Management (/portfolio/edit/:id)**

- My Portfolios page se jab user **Edit** par click karta hai, toh yeh full tabbed editor page khulta hai.
- Backend endpoint `GET /api/portfolios/:id` se user ka saved portfolio data load hota hai.
- User kisi bhi section ko edit kar sakta hai, skills add/remove kar sakta hai, aur projects/experience update kar sakta hai.
- **Theme & Visibility Tab:** Yahan se portfolio ka DaisyUI theme choose kar sakte hain aur Public/Private toggle kar sakte hain.
- **Live Preview Tab:** Real-time interactive portfolio preview dekh sakte hain.
- **Save Changes:** `PUT /api/portfolios/:id` request ke zariye saare changes MongoDB mein persist ho jaate hain.

**Step 5: Public vs Private Viewing Security System (/p/:slug)**

- Jab koi URL `/p/aarav-sharma-x7k2` kholta hai, toh request backend `GET /api/portfolios/public/:slug` par jati hai.
- **Public Portfolio (isPublic: true):** Koi bhi anonymous visitor bina login kiye portfolio dekh sakta hai.

- **Private Portfolio (isPublic: false):** Backend request ke token ko decode karke check karta hai. Agar token portfolio banane wale creator ka hai, toh portfolio display ho jata hai aur upar badge dikhta hai: "🔒 *Private Mode | Visible only to you*".
- Agar visitor creator nahi hai, toh backend 403 Forbidden bhejta hai aur screen par lock icon ke saath private message display hota hai.

**Step 6: Website Themes & Settings (/settings)**

- Settings page par user website interface ka theme badal sakta hai (Light, Dark ya 17+ DaisyUI themes).
- `SettingsContext` root `<html>` element par `data-theme` set karta hai jisse poori website par selected theme instantly apply ho jati hai.
- Saath hi English/Hindi language change aur Password security management bhi yahan milta hai.

## 🌐 6. Backend REST API Endpoints Quick Reference

| HTTP Method | Endpoint | Auth Needed | Description |
|---|---|---|---|
| POST | /api/auth/register | No | Naya user register karta hai |
| POST | /api/auth/login | No | Login verify karke JWT token generate karta hai |
| GET | /api/users/me | Yes | Logged-in user ki profile details fetch karta hai |
| PUT | /api/users/change-password | Yes | Current password verify karke new password set karta hai |
| GET | /api/portfolios | Yes | User ke banaye sabhi portfolios return karta hai |
| POST | /api/portfolios | Yes | Naya portfolio create aur publish karta hai |
| GET | /api/portfolios/:id | Yes | Portfolio edit karne ke liye fetch karta hai |
| PUT | /api/portfolios/:id | Yes | Existing portfolio ko update karta hai |
| DELETE | /api/portfolios/:id | Yes | Portfolio ko database se delete karta hai |
| GET | /api/portfolios/public/:slug | Conditional | Public view ya Private owner verification ke sath portfolio render karta hai |